

Intelligent Transportation Systems Integrating IoT and Machine Learning for Enhanced Urban Traffic Management

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Abstract

Traffic jam is a growing problem in urban areas, causing a delay, increased fuel usage, and environmental pollution. Accurate prediction of traffic conditions is essential for efficient transportation planning and management. This research aims to develop a traffic flow prediction model using techniques from Machine Learning . The proposed system will analyze historical traffic data of Delhi to predict future traffic patterns using models such as Linear Regression , time series graphs and Long Short-Term Memory (LSTM). The outcome of this research will aim to support intelligent decision-making in Intelligent Transportation Systems and contribute to urban development.[1], [2], [3]

Keywords

Traffic Prediction, LSTM, Regression, Time Series, Intelligent Transportation Systems, New Delhi

1. Introduction

Rapid urbanization and population growth have resulted in severe traffic jams and problem in major cities. New Delhi, being one of the busiest metropolitan areas, experiences unstable traffic patterns influenced by time, location, and human activity.

Traditional traffic prediction techniques often fail to capture temporal variations effectively. So, Accurate traffic flow prediction is a key component of ITS, as it enables the predication of future road conditions and supports proactive decision-making. Recent advancements in machine learning particularly LSTM networks, provide promising solutions by learning sequential patterns in time-series data. This research proposes a real-time traffic prediction and visualization system combining regression models and LSTM to enhance prediction accuracy and interpretability.

2. Related Work

2.1 Traffic Prediction

Traffic prediction has gained significant attention in recent years due to increasing urbanization and the rapid growth of transportation systems in metropolitan cities such as New Delhi. Accurate prediction of traffic conditions is essential for efficient traffic management and planning.[4]

Early research in traffic prediction mainly focused on statistical methods. These models are effective for capturing linear trends in time-series data but struggle to model non-linear and complex traffic patterns.[3], [5]

Further studies introduced traditional machine learning approaches such as linear regression. These models provide faster computation and are easy to implement. Regression models, in particular, are widely used due to their simplicity and ability to handle structured data. However, these methods fail to capture temporal dependencies since they treat observations independently.

In recent years Long Short-Term Memory (LSTM) networks have been widely adopted for traffic prediction. LSTM models are capable of learning long-term dependencies and temporal patterns in sequential data, making them highly effective for time-series forecasting. Despite their advantages, LSTM models require large datasets, higher computational resources, and longer training times.

Therefore, traditional models such as regression remain relevant due to their computational efficiency, while models like LSTM provide higher prediction accuracy. This study combines both regression and LSTM approaches to evaluate their performance and effectiveness in real-time traffic prediction.[2], [3]

2.2 Dataset Description

The dataset used in this study represents traffic conditions in New Delhi. It includes:[4]

- Timestamp (date and time)
- Traffic speed (continuous variable)
- Latitude and longitude (geospatial data)

3. Problem Statement

Rapid urban growth in metropolitan cities such as New Delhi has led to severe traffic congestion problem, resulting in increased travel time, fuel consumption, and environmental pollution. Existing traffic management approaches are largely reactive and fail to adapt to dynamic traffic conditions. Moreover, the large volume and continuous generation of traffic data make manual analysis impractical. Therefore, there is a need for an automated and efficient traffic prediction system capable of accurately forecasting traffic conditions in real time.[2], [4], [5]

4. Methodology

4.1 Data Preprocessing

Traffic data collected from metropolitan regions such as New Delhi is processed to improve model performance. The preprocessing steps include:

- Removal of missing and duplicate values
- Conversion of timestamp into datetime format
- Extraction of temporal features (hour, day)
- Normalization using MinMax scaling

MinMax Normalization Formula:

$$X_{norm} = \frac{X - X_{min}}{X_{max} - X_{min}}$$

Where:

- X = original value
- X_{min}, X_{max} = minimum and maximum values

4.2 Regression Models

Linear Regression

Linear regression models the relationship between input features and traffic speed.

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2$$

Where:

- y = predicted traffic speed
- x_1, x_2 = features (hour, day)

- β = model coefficients

Cost Function (Mean Squared Error):

$$J(\beta) = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

4.3 LSTM Model

LSTM is a recurrent neural network designed for time-series prediction.

A. LSTM Equations

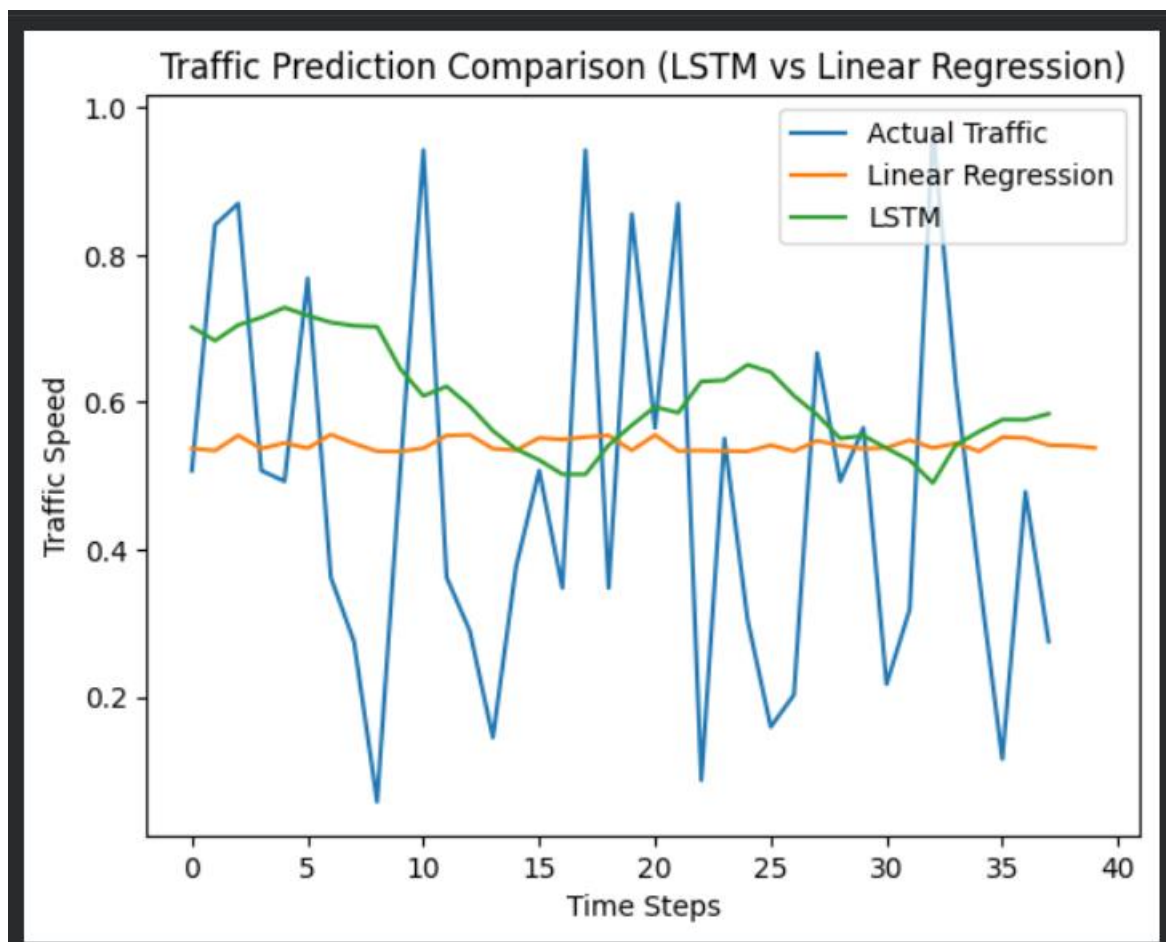
At each time step t :

Hidden State:

$$h_t = o_t \cdot \tanh(C_t)$$

Prediction Output

$$\hat{y}_t = W_y h_t + b_y$$



5. Model Training and Testing[3], [4], [5]

5.1 Data Splitting

To evaluate the performance of the proposed models, the dataset is divided into two parts:

- **Training Data (80%):** Used to train the models and learn patterns in traffic data.
- **Testing Data (20%):** Used to evaluate model performance on unseen data.

5.2 Training Phase

In the training phase, the models are trained on 80% of the dataset to learn traffic patterns.

- The models learn relationships between temporal features such as hour of the day and day of the week and the corresponding traffic conditions.
- Regression models learn how traffic speed changes with time.
- Classification models learn to categorize traffic into different levels such as low, medium, and high class.
- The LSTM model learns sequential dependencies, such as peak traffic hours and daily traffic trends.

5.3 Testing Phase

After training, the models are evaluated using the testing dataset to assess their performance on unseen data.

Process:

- The trained models are applied to the testing data.
- Predictions are generated for traffic conditions.

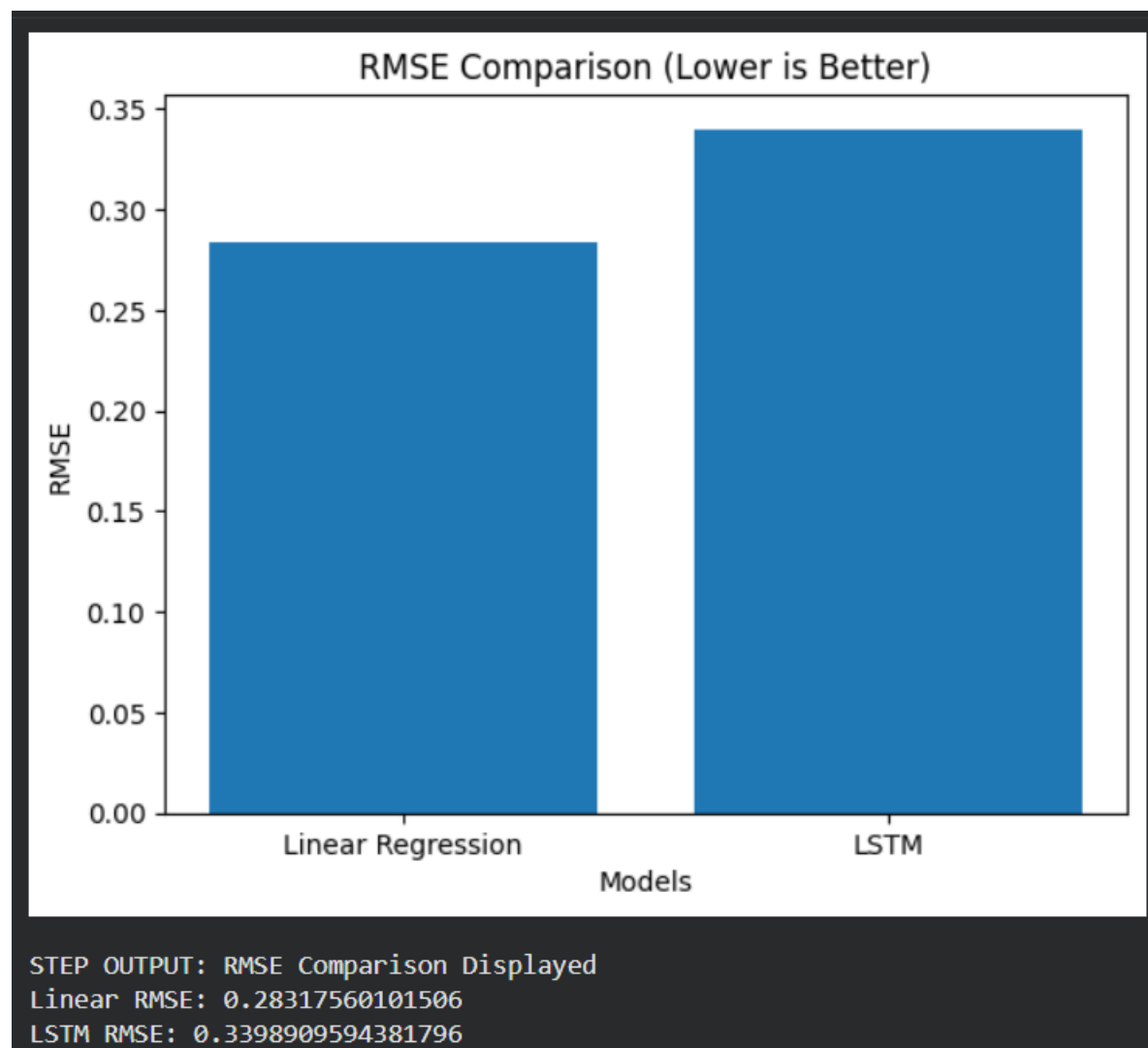
Model Outputs:

- **Linear Regression** → Predicts continuous traffic speed value
- **LSTM** → Predicts future traffic values using time-series sequences

Root Mean Squared Error (RMSE):

$$RMSE = \sqrt{\frac{1}{n} \sum (y - \hat{y})^2}$$

Lower RMSE values indicate better model performance.



6. Dataset

This research is based on a traffic dataset representing conditions in New Delhi. The dataset includes time-based and geographical information related to traffic flow.

Dataset Description

- Traffic data includes:
 - Timestamp (date and time)
 - Traffic speed values
 - Latitude and longitude coordinates

Divide this dataset into 80:20 ratio.

- Training data = 80% (approx 200)
- Testing data = 20% (approx 40)

7. Results Analysis

A. Experimental Results

The proposed model Linear Regression, Long Short-Term Memory (LSTM) were trained and evaluated on the traffic dataset using an 80:20 train–test split. Performance was assessed using both root mean square error(RMSE) and accuracy.

Linear Regression, used as a baseline, achieved the low performance due to its assumption of linear relationships, which is insufficient for capturing the complex and dynamic nature of traffic data.

The LSTM model achieving the lower RMSE and high accuracy. This is attributed to its ability to learn temporal dependencies and sequential patterns, such as peak-hour congestion and daily traffic trends.

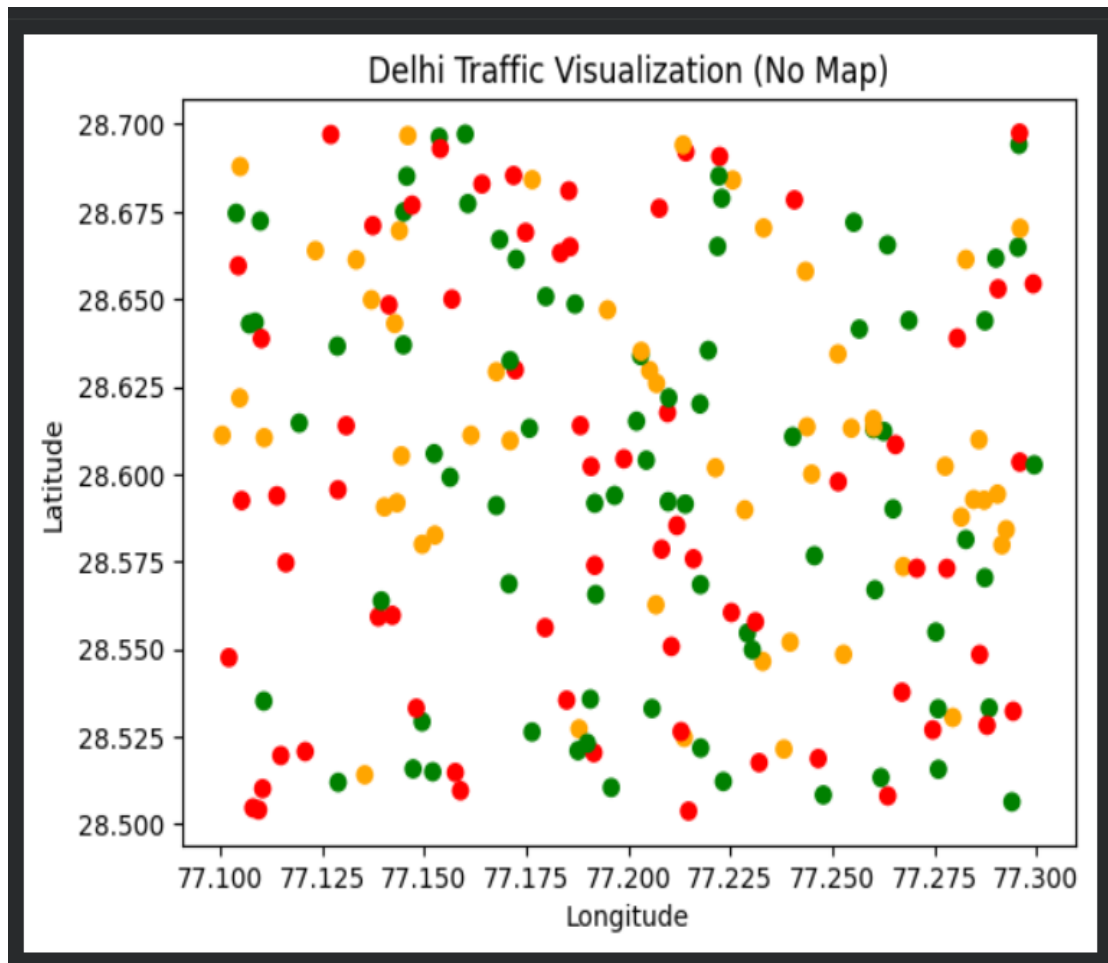
B. Graphical Analysis

The comparison graph between actual and predicted traffic values shows that:

- Linear Regression produces smoother predictions that fail to follow rapid fluctuations in traffic patterns.
- LSTM predictions closely align with actual traffic values, indicating superior learning of time-dependent behavior.

Traffic conditions are categorized into three classes:

- **Class 0:** Low traffic
- **Class 1:** Medium traffic
- **Class 2:** High traffic



9. Future Scope

In the future, the proposed traffic prediction system can be further enhanced by incorporating more advanced machine learning and deep learning techniques such as Support Vector Machines (SVM), Random Forest, and transformer-based models

The model can also be extended to include real-time data integration using Internet of Things (IoT) sensors, GPS data, and live traffic feeds. This would enable dynamic and real-time traffic prediction for better decision-making. Furthermore, incorporating additional features such as weather conditions, road incidents, and public events can significantly improve model performance.[1], [2], [3]

10. Conclusion

This study proposed a traffic prediction system using machine learning techniques, including Linear Regression, Long Short-Term Memory (LSTM) models. The results demonstrate that traditional models such as Linear Regression provide a simple and computationally efficient baseline but are limited in capturing complex traffic patterns. However, the LSTM model achieved the best results in terms of accuracy and error reduction due to its ability to learn temporal dependencies in time-series traffic data.[1], [3]

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